# Shaurya Baranwal

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## Education

**Boston University** *Aug 2026 - May 2028*
*MS in Computer Science and Aritifical Intelligence (Online)*

**Rutgers University - New Brunswick** *Sept 2022 – May 2026*
*BS in Computer Science and Mathematics, Certificate in Mathematical Finance*

- **Honors:** Rutgers Honors College, Phi Beta Kappa, Dean's List

## Experience

**Software Engineer Intern** *Remote*
*Beunec Technologies* *May 2025 – Aug 2025*

- Developed and integrated platform features like Neo Links, Beunec Docs, a rich online word processor that supports live collaboration, real-time editing, and document sharing, and Task Flow, a task management system with real-time updates.
- Optimized real-time database performance and implemented secure authentication systems using Cloudflare Workers and Auth0.

**Course Assistant for Internet Technology** *New Brunswick, NJ*
*Department of Computer Science - Rutgers University* *Sept 2025 – Jan 2026*

- Graded and provided feedback on exams and Wireshark labs for over 150 students in a junior-level Computer Networks course, ensuring fairness and accuracy.
- Held weekly office hours to assist students with concepts such as TCP congestion control, routing algorithms, and packet analysis using Wireshark.

**Research Assistant** *New Brunswick, NJ*
*Department of Computer Science - Rutgers University* *Sept 2025 – Feb 2026*

- Designed and implemented a QUBO-based Support Vector Machine (SVM) pipeline, converting the SVM optimization objective into a QUBO matrix and solving it using NASA's PySA quantum-inspired annealing framework.
- Collaborated with Professor Minsung Kim to extend PySA's Ising and RBM tutorials into a novel application: QUBO-based SVM classification.

**Math Grader** *New Brunswick, NJ*
*Department of Mathematics - Rutgers University* *Sept 2024 – Dec 2024*

- Graded and provided feedback on weekly homework assignments for over 150 students in "Intro to Linear Algebra".

## Projects

**Deep Q-Learning Geometry Dash Agent** *Feb 2026 - Mar 2026*

- Built an autonomous real-time gaming agent in **PyTorch** using a **Dueling Deep Q-Network (Dueling DQN)** with dual value/advantage streams to master dynamic platformer stages.
- Implemented a low-latency perception loop via **mss** and **OpenCV**, processing 4-frame grayscale histories ($128 \times 128$) with **Prioritized Experience Replay (PER)** and multi-threaded background optimization.
- Engineered an automated menu navigation bot and progressive curriculum pipeline, utilizing template matching for state transitions and hardware-timed inference (30 FPS) to prevent scheduler drift.

**Personalized Chatbot** *January 2026*

- Developed a **Python**-based **FastAPI** backend to orchestrate vector retrieval, integrating it with a **Next.js** and **TypeScript** interface for seamless user interaction.
- Built a high-performance search engine using **Pinecone** and **OpenAI embeddings**, utilizing the **Vercel AI SDK** to deliver low-latency, streaming model outputs.

## Skills

**Languages:** Python, Java, C/C++, Javascript, Typescript, OCaml, HTML, CSS, Maple, SQL

**Frameworks/Libraries:** React.js, React Native, Hono.js, PyTorch, TensorFlow, Flask, FastAPI, Next.js, Express.js, Node.js, Pandas, Numpy, LangChain

**Other Tools:** Git, Firebase, Jupyter Notebook, Linux, Clerk, Postgres, Cloudflare, Docker